

Repetitive rectal painful distention induces rectal hypersensitivity in patients with irritable bowel syndrome.

BACKGROUND: A reduced rectal perceptual threshold has been reported in patients with irritable bowel syndrome (IBS), but this phenomenon may be induced by a comorbid psychological state. We evaluated the rectal pain threshold at baseline and after conditioning (repetitive rectal painful distention: RRD) in patients with IBS or functional abdominal pain syndrome (FAPS), which is an abdominal pain disorder, and in healthy controls, and determined whether rectal hypersensitivity is a reliable marker for IBS. **METHODS:** The rectal sensory threshold was assessed by a barostat. First, a ramp distention of 40 ml/min was induced, and the threshold of pain and the maximum tolerable pressure (mmHg) were measured. Next, RRD (phasic distentions of 60-s duration separated by 30-s intervals) was given with a tracking method until the subjects had complained of pain six times. Finally, ramp distention was induced again, and the same parameters were measured. The normal value was defined by calculating the 95% confidence intervals of controls. **RESULTS:** Five or six of the seven IBS patients showed a reduced rectal pain threshold or maximum tolerable pressure, respectively, at baseline. In all patients with IBS, both thresholds were reduced after RRD load, but they were reduced in none of the patients with FAPS. RRD significantly reduced both thresholds in the IBS group ($P < 0.05$), but it had no effect in the control or FAPS groups. **CONCLUSIONS:** Rectal hypersensitivity induced by RRD may be a reliable marker for IBS. Conditioning-induced visceral hypersensitivity may play a pathophysiologic role in IBS.